

EC601 — Control System

Semester VI • 3L:0T:0P • 3 Credits

Module 4 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Frequency Response Analysis — Introduction

Frequency response analysis studies how a system responds to sinusoidal inputs of varying frequency. For a stable LTI system with transfer function $G(j\omega)$, a sinusoidal input $r(t) = A \sin(\omega t)$ produces a steady-state output $c(t) = A|G(j\omega)| \sin(\omega t + \angle G(j\omega))$.

Frequency Response Function: Obtained by substituting $s = j\omega$ in the transfer function:

$$G(j\omega) = G(s)|_{s=j\omega} = |G(j\omega)| \angle \phi(\omega)$$

Magnitude: $|G(j\omega)|$ — ratio of output to input amplitude.

Phase: $\phi(\omega) = \angle G(j\omega)$ — phase shift of output relative to input.

Plot Type	Description
Polar Plot	Locus of $G(j\omega)$ in the complex plane as $\omega : 0 \rightarrow \infty$. Also called Nyquist plot (open-loop).
Bode Plot	Two separate plots: $ G(j\omega) $ in dB vs $\log \omega$, and $\angle G(j\omega)$ in degrees vs $\log \omega$.
Nyquist Plot	Polar plot of $G(j\omega)H(j\omega)$ for $\omega : -\infty \rightarrow +\infty$; used for Nyquist stability criterion.
Nichols Chart	Magnitude vs phase plot; used for closed-loop frequency response from open-loop data.

2. Polar Plots

Polar Plot: A plot of $G(j\omega)$ in the complex (Re–Im) plane as ω varies from 0 to $+\infty$. Each point on the curve represents the tip of the phasor $G(j\omega)$ at a particular frequency. The distance from the origin is $|G(j\omega)|$ and the angle is $\angle G(j\omega)$.

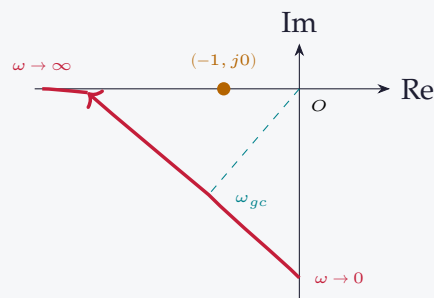
2.1 Polar Plot of Standard Factors

Factor	At $\omega = 0$	At $\omega \rightarrow \infty$
K (gain)	$(K, 0^\circ)$	$(K, 0^\circ)$ — point on real axis
$j\omega$ (integrator)	$\infty \angle -90^\circ$	$0 \angle -90^\circ$ — negative imaginary axis
$1/j\omega$	$\infty \angle -90^\circ$	$0 \angle -90^\circ$ — negative imaginary axis
$1/(1 + j\omega T)$	$(1, 0^\circ)$	$(0, -90^\circ)$ — 4th quadrant semicircle
$1/(j\omega)^2$	$\infty \angle -180^\circ$	$0 \angle -180^\circ$ — negative real axis
$1/[j\omega(1 + j\omega T)]$	$\infty \angle -90^\circ$	$0 \angle -180^\circ$ — 3rd quadrant

2.2 Key Properties of Polar Plots

- The **starting point** ($\omega = 0$) and **ending point** ($\omega \rightarrow \infty$) are determined by the system type and order.
- For a **Type 0** system: starts on positive real axis; for **Type 1**: starts at -90° infinity; for **Type 2**: starts at -180° infinity.
- The **number of clockwise encirclements** of the $(-1, j0)$ point determines closed-loop stability (Nyquist criterion).
- The **gain crossover frequency** ω_{gc} : where $|G(j\omega)| = 1$ (0 dB).
- The **phase crossover frequency** ω_{pc} : where $\angle G(j\omega) = -180^\circ$.

Polar Plot — Type 1, Second Order: $G(j\omega) = K/[j\omega(1 + j\omega T)]$



3. Bode Plot

Bode Plot: Two semi-log plots of the open-loop frequency response $G(j\omega)$:

- **Magnitude plot:** $20 \log_{10} |G(j\omega)|$ in dB vs $\log_{10} \omega$ (log scale).
- **Phase plot:** $\angle G(j\omega)$ in degrees vs $\log_{10} \omega$ (log scale).

Bode plots use **asymptotic approximations** (straight-line segments) for quick

sketching.

3.1 Standard Factors and Their Bode Contributions

Factor	Magnitude (dB)	Phase
Gain K	$20 \log K$ (constant)	0° (if $K > 0$); -180° (if $K < 0$)
$j\omega$ (zero at origin)	$+20$ dB/dec, passes through 0 dB at $\omega = 1$	$+90^\circ$ (constant)
$1/j\omega$ (pole at origin)	-20 dB/dec, passes through 0 dB at $\omega = 1$	-90° (constant)
$(1 + j\omega T)$ (simple zero)	0 dB for $\omega \ll 1/T$; $+20$ dB/dec for $\omega \gg 1/T$	$0^\circ \rightarrow +90^\circ$, 45° at $\omega = 1/T$
$1/(1 + j\omega T)$ (simple pole)	0 dB for $\omega \ll 1/T$; -20 dB/dec for $\omega \gg 1/T$	$0^\circ \rightarrow -90^\circ$, -45° at $\omega = 1/T$
$1/(j\omega)^2$ (double pole)	-40 dB/dec	-180° (constant)
2nd order factor	0 dB below ω_n ; -40 dB/dec above ω_n ; resonance peak near ω_n	$0^\circ \rightarrow -180^\circ$, -90° at ω_n

3.2 Corner Frequency (Break Frequency)

Corner Frequency

For a factor $(1 + j\omega T)$: **corner frequency** $\omega_c = 1/T$ rad/s.

Asymptotic error at corner frequency:

- Magnitude: actual value differs from asymptote by ± 3 dB at $\omega = \omega_c$.
- Phase: actual = $\pm 45^\circ$ at ω_c ; asymptote approximation has $\pm 5.7^\circ$ max error.

3.3 Bode Plot Sketching Procedure

- Step 1:** Write $G(j\omega)$ in **time-constant form**: factor out constants, express each factor as $(1 + j\omega T)$.
- Step 2:** Compute the **initial slope**: $-20N$ dB/dec where N = system type (number of poles at origin).
- Step 3:** Mark all **corner frequencies** $\omega_c = 1/T$ on the ω -axis.
- Step 4:** Draw the **magnitude asymptotes**: slope changes by $+20$ dB/dec at each zero corner, -20 dB/dec at each pole corner.
- Step 5:** Apply **3 dB corrections** at each corner frequency if accuracy is needed.
- Step 6:** Draw the **phase plot**: sum contributions from each factor; use $\pm 45^\circ$ /decade approximation.

Example: $G(j\omega) = \frac{10}{j\omega(1 + 0.1j\omega)(1 + 0.5j\omega)}$

Factors: Gain $K = 10$ (+20 dB), pole at origin (−20 dB/dec from start), poles at $\omega_c = 10$ and $\omega_c = 2$.

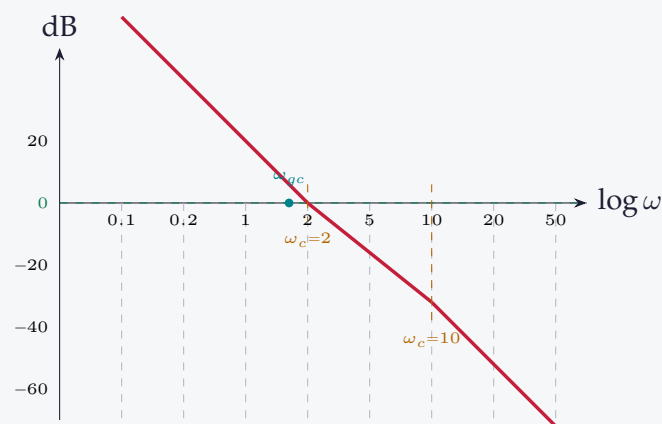
Initial value: At $\omega = 1$: $20 \log(10/1) = 20$ dB (from gain and pole at origin).

Slope changes:

- $\omega < 2$: slope = −20 dB/dec (one pole at origin)
- $2 < \omega < 10$: slope = −40 dB/dec (pole at $\omega = 2$ adds −20)
- $\omega > 10$: slope = −60 dB/dec (pole at $\omega = 10$ adds another −20)

Phase: -90° (pole at origin) $-\tan^{-1}(0.5\omega) - \tan^{-1}(0.1\omega)$

Bode Magnitude Plot Sketch



4. Stability in Frequency Domain

4.1 Gain Margin (GM)

Gain Margin: The factor by which the open-loop gain can be *increased* before the closed-loop system becomes unstable. Measured at the **phase crossover frequency** ω_{pc} (where $\angle G(j\omega)H(j\omega) = -180^\circ$).

$$GM = \frac{1}{|G(j\omega_{pc})H(j\omega_{pc})|}$$

In dB: $GM_{dB} = -20 \log_{10} |G(j\omega_{pc})H(j\omega_{pc})|$

Stable system: $GM > 1$ (i.e., $GM_{dB} > 0$ dB). **Unstable:** $GM < 1$ ($GM_{dB} < 0$ dB).

4.2 Phase Margin (PM)

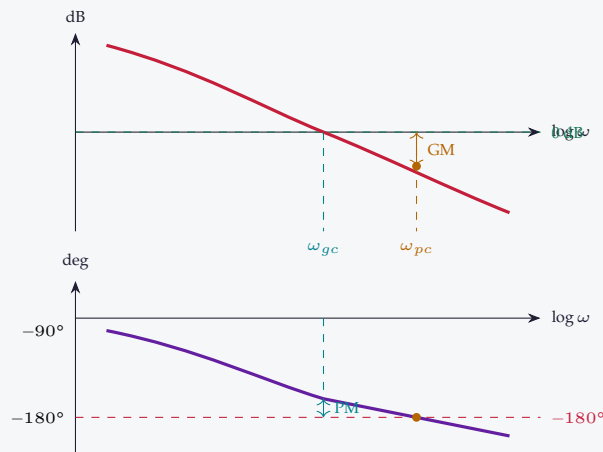
Phase Margin: The additional phase lag that can be introduced before the closed-loop system becomes unstable. Measured at the **gain crossover frequency** ω_{gc} (where $|G(j\omega)H(j\omega)| = 1$, i.e., 0 dB).

$$PM = 180^\circ + \angle G(j\omega_{gc})H(j\omega_{gc})$$

Stable system: $PM > 0^\circ$. **Unstable:** $PM < 0^\circ$.

Typical design target: $GM \geq 6$ dB, $PM \geq 30^\circ$ – 60° .

GM and PM on Bode Plot



Margin	Definition	Measurement Point
Gain Margin (GM)	$1/ G(j\omega_{pc}) $	At phase crossover freq ω_{pc} ($\angle = -180^\circ$)
Phase Margin (PM)	$180^\circ + \angle G(j\omega_{gc})$	At gain crossover freq ω_{gc} ($ G = 1$)
Stable condition	$GM > 1$ (> 0 dB)	$PM > 0^\circ$
Good design	$GM \geq 6$ dB	$PM = 30^\circ$ – 60°

Exam Tips — GM and PM

- **GM** is read from the magnitude plot at ω_{pc} ; **PM** is read from the phase plot at ω_{gc} .
- If the magnitude curve *never* crosses 0 dB, ω_{gc} does not exist — system is always stable for that gain.
- If the phase curve *never* reaches -180° , ω_{pc} does not exist — $GM = \infty$.
- **Higher PM** $\Rightarrow$ less overshoot, more damping. $PM \approx 100\zeta$ for $\zeta < 0.7$.
- **Conditionally stable systems** can have $GM > 0$ dB but become unstable if gain is *reduced*.

5. Nyquist Plot

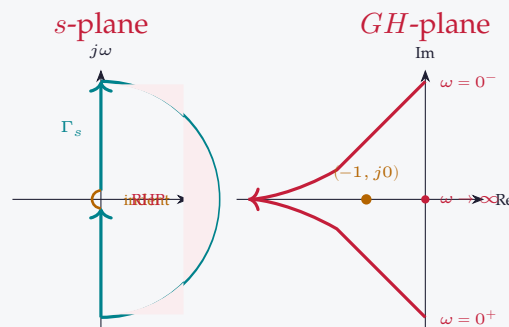
Nyquist Plot: The polar plot of $G(j\omega)H(j\omega)$ for ω varying from $-\infty$ to $+\infty$ (i.e., the complete Nyquist contour). It is the mirror image of the polar plot about the real axis for negative frequencies. Used to apply the **Nyquist Stability Criterion**.

5.1 Nyquist Contour

The **Nyquist contour** Γ_s in the s -plane encloses the entire right half-plane (RHP):

- Segment 1: $s = j\omega, \omega : 0^+ \rightarrow +\infty$ (positive imaginary axis)
- Segment 2: $s = Re^{j\theta}, R \rightarrow \infty, \theta : +90^\circ \rightarrow -90^\circ$ (large semicircle)
- Segment 3: $s = j\omega, \omega : -\infty \rightarrow 0^-$ (negative imaginary axis)
- **Indentation:** Small semicircle of radius $\varepsilon \rightarrow 0$ around any poles on the imaginary axis.

Nyquist Contour in s -plane and Corresponding $G(j\omega)H(j\omega)$ Plot



6. Nyquist Stability Criterion

Nyquist Stability Criterion: Based on the **Principle of the Argument** (Cauchy's theorem). For a closed-loop system with open-loop transfer function $G(s)H(s)$:

$$N = Z - P$$

where:

- N = number of **clockwise encirclements** of the $(-1, j0)$ point by the Nyquist plot of $G(j\omega)H(j\omega)$.
- Z = number of **closed-loop poles** in the RHP (zeros of $1 + GH$).
- P = number of **open-loop poles** of $G(s)H(s)$ in the RHP.

For stability: $Z = 0$, therefore $N = -P$ (i.e., P counter-clockwise encirclements if $P \neq 0$; no encirclements if $P = 0$).

6.1 Applying the Nyquist Criterion — Step by Step

- Step 1:** Determine P = number of open-loop poles of $G(s)H(s)$ in the RHP.
- Step 2:** Sketch the Nyquist plot of $G(j\omega)H(j\omega)$ for $\omega : 0^+ \rightarrow +\infty$.
- Step 3:** Mirror the plot about the real axis for $\omega : -\infty \rightarrow 0^-$.
- Step 4:** Handle poles on the $j\omega$ -axis by indenting the contour (small semicircle of radius $\varepsilon \rightarrow 0$).
- Step 5:** Count N = net clockwise encirclements of $(-1, j0)$.
- Step 6:** Compute $Z = N + P$. If $Z = 0$: **stable**. If $Z > 0$: **unstable** (Z RHP poles).

Example: $G(s)H(s) = K/[s(s+1)(s+2)]$, find stability range

Open-loop poles: $s = 0$ (on $j\omega$ -axis), $s = -1$, $s = -2$ (all in LHP). So $P = 0$.

Nyquist plot crosses the negative real axis at ω_{pc} where $\angle G(j\omega)H(j\omega) = -180^\circ$.

At ω_{pc} : $\angle = -90^\circ - \tan^{-1}(\omega) - \tan^{-1}(\omega/2) = -180^\circ \Rightarrow \tan^{-1}(\omega) + \tan^{-1}(\omega/2) = 90^\circ$
 $\Rightarrow \omega_{pc} = \sqrt{2}$ rad/s.

$$|G(j\sqrt{2})H(j\sqrt{2})| = \frac{K}{\sqrt{2} \cdot \sqrt{3} \cdot \sqrt{4}} = \frac{K}{2\sqrt{6}}$$

For stability: $N = 0$ requires the plot does not encircle $(-1, j0)$:

$$\frac{K}{2\sqrt{6}} < 1 \Rightarrow K < 2\sqrt{6} \approx 4.9$$

Gain Margin: $GM = 2\sqrt{6}/K$. For $K = 1$: $GM = 2\sqrt{6} \approx 4.9$ (13.8 dB).

6.2 Simplified Nyquist Criterion (Minimum Phase Systems)

For Minimum Phase Systems ($P = 0$)

A minimum phase system (no open-loop RHP poles or zeros) is **closed-loop stable** if and only if the Nyquist plot of $G(j\omega)H(j\omega)$ does **not encircle** the $(-1, j0)$ point.

Equivalently (from Bode plot): **stable** if $|G(j\omega_{pc})H(j\omega_{pc})| < 1$, i.e., $GM > 0$ dB AND $PM > 0^\circ$.

Exam Tips — Nyquist Criterion

- $N = Z - P$ — memorise this. $Z = 0$ for stability.
- For **minimum phase** systems ($P = 0$): stable iff no encirclement of $(-1, j0)$.
- **Clockwise** encirclement = positive N ; **counter-clockwise** = negative N .
- Poles on $j\omega$ -axis (e.g., integrators): indent contour to the right; the large semicircle maps to origin.
- The Nyquist plot is the **mirror image** of the polar plot about the real axis.
- $(-1, j0)$ **point** is the critical point — corresponds to $|GH| = 1$ and $\angle GH = -180^\circ$.

7. Performance Specifications in Frequency Domain

Frequency Domain Specifications describe the closed-loop system's behaviour in terms of its frequency response $T(j\omega) = C(j\omega)/R(j\omega)$. They are directly related to time-domain performance.

7.1 Closed-Loop Frequency Response Specifications

Specification	Symbol	Definition
Resonant Peak	M_r	Maximum value of $ T(j\omega) $. Indicates relative stability; high M_r means low damping.
Resonant Frequency	ω_r	Frequency at which $ T(j\omega) $ is maximum.
Bandwidth	ω_{BW}	Frequency at which $ T(j\omega) $ drops to $1/\sqrt{2}$ (−3 dB) of its DC value. Indicates speed of response.
Cutoff Rate	—	Rate at which $ T(j\omega) $ decreases beyond ω_{BW} ; indicates noise rejection.

7.2 Formulas for Second-Order System

Closed-Loop Frequency Response — 2nd Order

For $T(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$:

$$|T(j\omega)| = \frac{1}{\sqrt{(1 - u^2)^2 + (2\zeta u)^2}}, \quad u = \omega/\omega_n$$

Resonant frequency: $\omega_r = \omega_n \sqrt{1 - 2\zeta^2}$ (exists only for $\zeta < 1/\sqrt{2} \approx 0.707$)

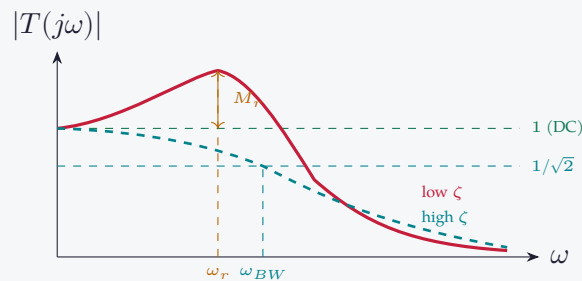
Resonant peak: $M_r = \frac{1}{2\zeta\sqrt{1 - \zeta^2}}$ (for $\zeta < 0.707$)

Bandwidth: $\omega_{BW} = \omega_n \sqrt{(1 - 2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$

7.3 Relation Between Frequency and Time Domain Specs

Freq. Domain	Relation	Time Domain Equivalent
High ω_{BW}	$\omega_{BW} \propto \omega_n$	Faster rise time, faster response
High M_r	$M_r \uparrow$ as $\zeta \downarrow$	High overshoot M_p
$M_r = 1$	$\zeta \geq 0.707$	No overshoot (overdamped/critically)
Phase Margin	$PM \approx 100\zeta$ (for $\zeta < 0.7$)	Damping ratio ζ
Gain Margin	Larger GM	More stable, less sensitive to gain
ω_{gc}	$\omega_{gc} \approx \omega_{BW}$	Bandwidth approximation

Closed-Loop Frequency Response (Resonance Peak)



Exam Tips — Frequency Domain Specs

- $M_r > 1.3$ (about 2.3 dB) indicates poor relative stability (excessive overshoot).
- $\omega_r < \omega_n$ always (resonant frequency is less than natural frequency).
- $\omega_{BW} > \omega_n$ always (bandwidth exceeds natural frequency).
- $PM \approx 100\zeta$ for $\zeta < 0.7$ — very useful approximation in design.
- **Bandwidth** determines speed: wider BW = faster response but more noise.
- For $\zeta \geq 0.707$: $M_r = 1$ (no resonance peak); response is monotonically decreasing.

Quick Revision — Module 4

Frequency Response Analysis | EC601

Key Formulas at a Glance

Frequency response	$G(j\omega) = G(s) _{s=j\omega}$
Gain Margin	$GM = 1/ G(j\omega_{pc})H(j\omega_{pc}) $
GM in dB	$GM_{dB} = -20 \log G(j\omega_{pc}) $
Phase Margin	$PM = 180^\circ + \angle G(j\omega_{gc})H(j\omega_{gc})$
Phase crossover freq	ω_{pc} : where $\angle GH = -180^\circ$
Gain crossover freq	ω_{gc} : where $ GH = 1$ (0 dB)
Nyquist criterion	$N = Z - P$; stable if $Z = 0$
Resonant peak (2nd order)	$M_r = 1/(2\zeta\sqrt{1-\zeta^2})$, $\zeta < 0.707$
Resonant frequency	$\omega_r = \omega_n\sqrt{1-2\zeta^2}$
Bandwidth (2nd order)	$\omega_{BW} = \omega_n\sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$
PM–damping relation	$PM \approx 100\zeta$ (for $\zeta < 0.7$)
Bode: simple pole slope	-20 dB/dec above corner freq $\omega_c = 1/T$
Bode: double pole slope	-40 dB/dec above ω_n
Bode: phase at corner	-45° at ω_c for simple pole

One-Line Definitions

- **Frequency Response:** System's steady-state response to sinusoidal input; $G(j\omega)$.
- **Polar Plot:** Locus of $G(j\omega)$ in complex plane as $\omega : 0 \rightarrow \infty$.
- **Bode Plot:** Semi-log plots of magnitude (dB) and phase (deg) vs $\log \omega$.
- **Corner Frequency:** $\omega_c = 1/T$; slope changes by ± 20 dB/dec here.
- **Gain Crossover Freq ω_{gc} :** Where $|GH| = 1$ (0 dB); PM is measured here.
- **Phase Crossover Freq ω_{pc} :** Where $\angle GH = -180^\circ$; GM is measured here.
- **Gain Margin:** Factor by which gain can increase before instability; $GM > 0$ dB for stability.
- **Phase Margin:** Additional phase lag before instability; $PM > 0^\circ$ for stability.
- **Nyquist Plot:** Polar plot of $GH(j\omega)$ for $\omega : -\infty \rightarrow +\infty$.
- **Nyquist Criterion:** $N = Z - P$; stable if $Z = 0$ (no RHP closed-loop poles).

- **Resonant Peak M_r :** Maximum of $|T(j\omega)|$; indicates damping ($M_r \uparrow$ as $\zeta \downarrow$).
- **Bandwidth ω_{BW} :** Freq where $|T|$ drops to -3 dB; indicates response speed.
- **Minimum Phase System:** No RHP poles or zeros; $P = 0$; stable iff no encirclement of $(-1, j0)$.

Frequently Asked / Viva Questions

Q1. What is the frequency response of a system?

The steady-state output-to-input ratio for sinusoidal inputs; obtained by substituting $s = j\omega$ in $G(s)$.

Q2. What is the gain crossover frequency?

The frequency ω_{gc} where $|G(j\omega)H(j\omega)| = 1$ (0 dB). Phase margin is measured here.

Q3. What is the phase crossover frequency?

The frequency ω_{pc} where $\angle G(j\omega)H(j\omega) = -180^\circ$. Gain margin is measured here.

Q4. Define Gain Margin.

$GM = 1/|GH(j\omega_{pc})|$. It is the factor by which gain can be increased before instability. Stable if $GM > 1$ (0 dB).

Q5. Define Phase Margin.

$PM = 180^\circ + \angle GH(j\omega_{gc})$. Additional phase lag before instability. Stable if $PM > 0^\circ$.

Q6. State the Nyquist stability criterion.

$N = Z - P$. For stability $Z = 0$, so $N = -P$. For minimum phase ($P = 0$): no encirclement of $(-1, j0)$.

Q7. What is N, Z, P in Nyquist criterion?

N = clockwise encirclements of $(-1, j0)$; Z = RHP closed-loop poles; P = RHP open-loop poles.

Q8. What is the slope of Bode magnitude plot for a simple pole?

-20 dB/decade above the corner frequency $\omega_c = 1/T$.

Q9. What is the resonant peak M_r ?

Maximum value of closed-loop frequency response; $M_r = 1/(2\zeta\sqrt{1-\zeta^2})$ for 2nd order. Exists only for $\zeta < 0.707$.

Q10. What is bandwidth?

Frequency at which $|T(j\omega)|$ drops to $1/\sqrt{2}$ (-3 dB) of DC value. Wider BW = faster response.

Q11. Relation between PM and damping ratio?

$PM \approx 100\zeta$ for $\zeta < 0.7$. So $PM = 45^\circ$ corresponds to $\zeta \approx 0.45$.

Q12. What is a minimum phase system?

A system with no RHP poles or zeros and no time delay. Its Bode magnitude uniquely determines the phase.

Stability Summary

Condition	Stable	Unstable
Gain Margin	$GM > 1$ (> 0 dB)	$GM < 1$ (< 0 dB)
Phase Margin	$PM > 0^\circ$	$PM < 0^\circ$
Nyquist ($P = 0$)	No encirclement of $(-1, j0)$	Encircles $(-1, j0)$
Nyquist (general)	$Z = N + P = 0$	$Z = N + P > 0$
Bode (min. phase)	$\omega_{gc} < \omega_{pc}$	$\omega_{gc} > \omega_{pc}$